

Open Science in Practice: Collaborative Research with Git



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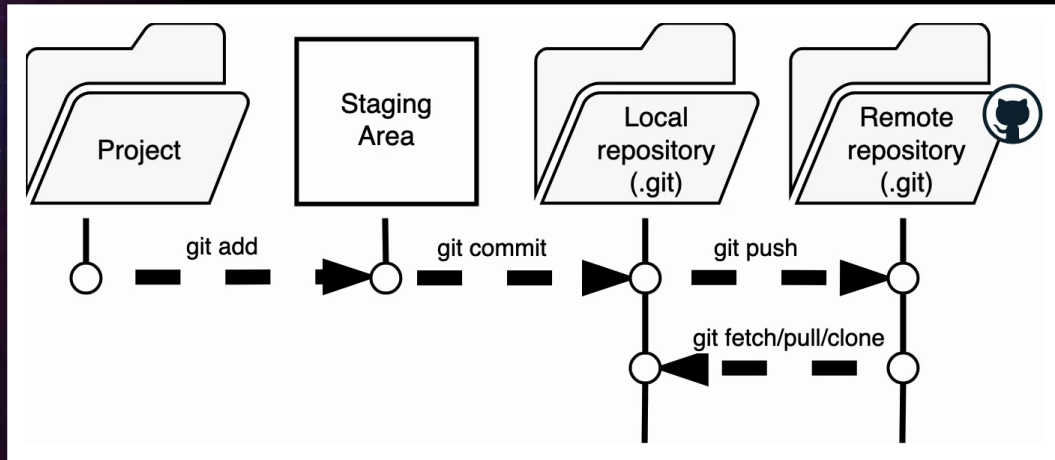
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Git collaboration

Andreas Gerhardsson

Recap

```
git add | git commit | git push  
git branch  
git pull
```



Git collaboration

What | When | Why | **By Whom!**



Cloning



`git clone`



- You can clone any repository from eg. GitHub
- A fresh start (including commit history)
- Work on it as your own (always check license...)
- Add your own remote
- No inherent link to original



Git course



`.git`



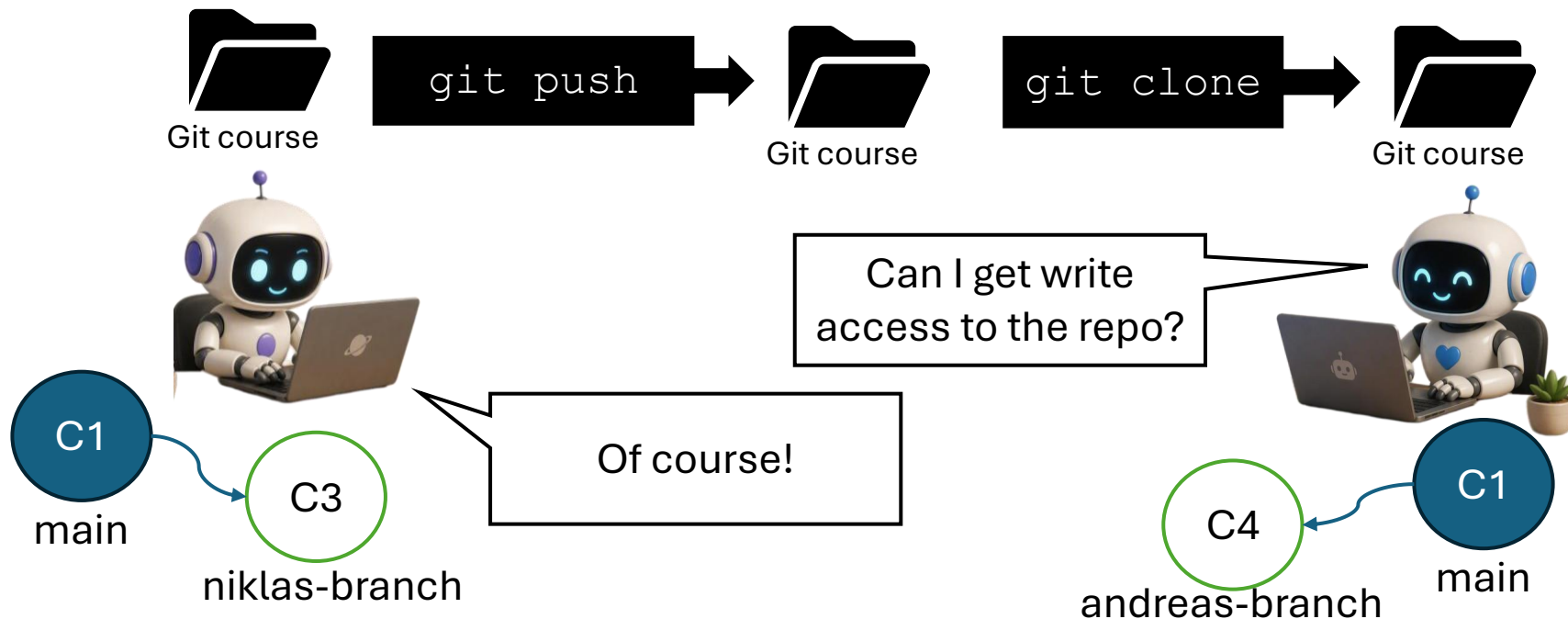
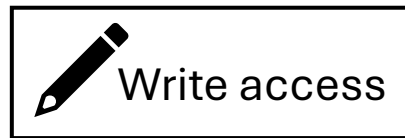
`read_data.R`



`plot_results.R`

```
andreas ~ % git remote set-upstream origin \  
git@github.com:k-CIR/course-open-  
science-in-practice.git
```

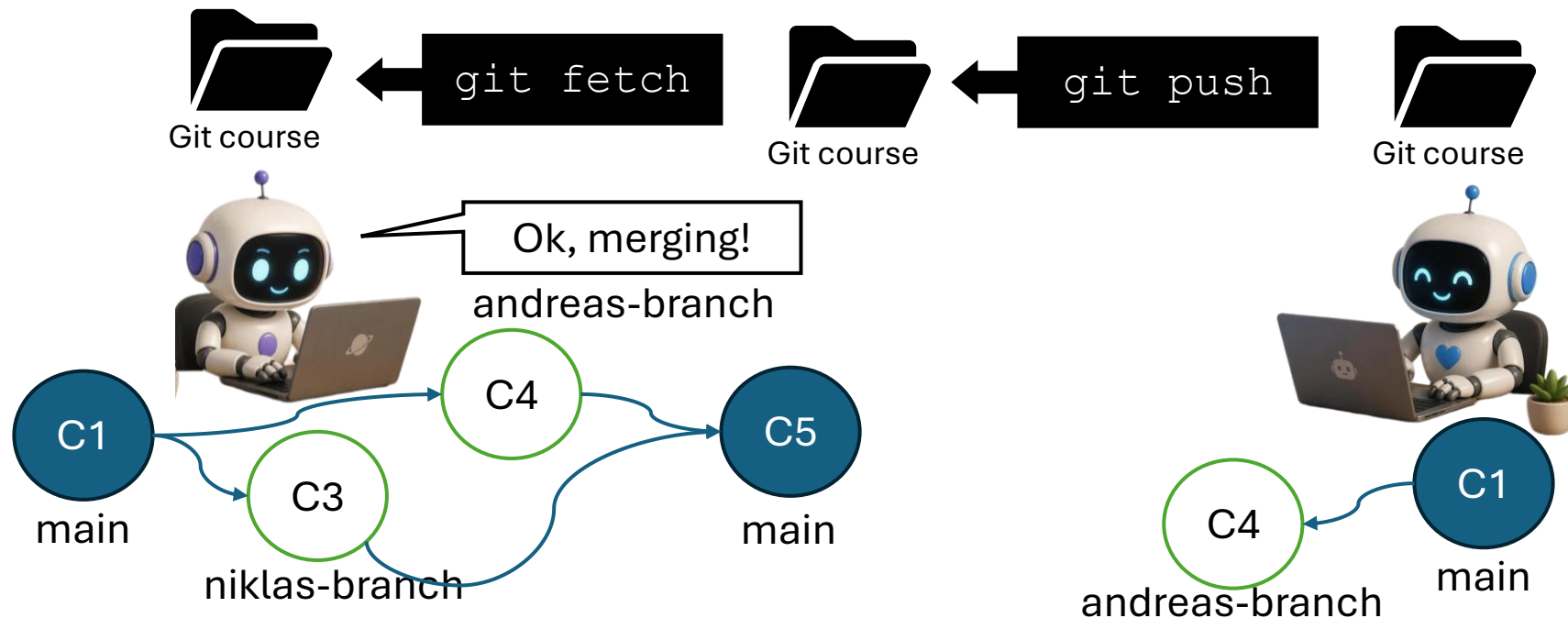
Sharing the same remote



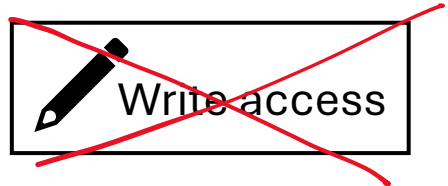
Sharing the same remote



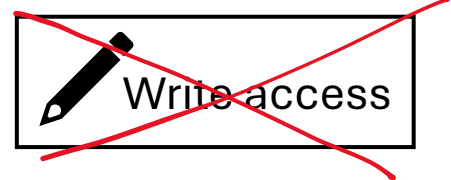
Write access



Forking a repository



Forking



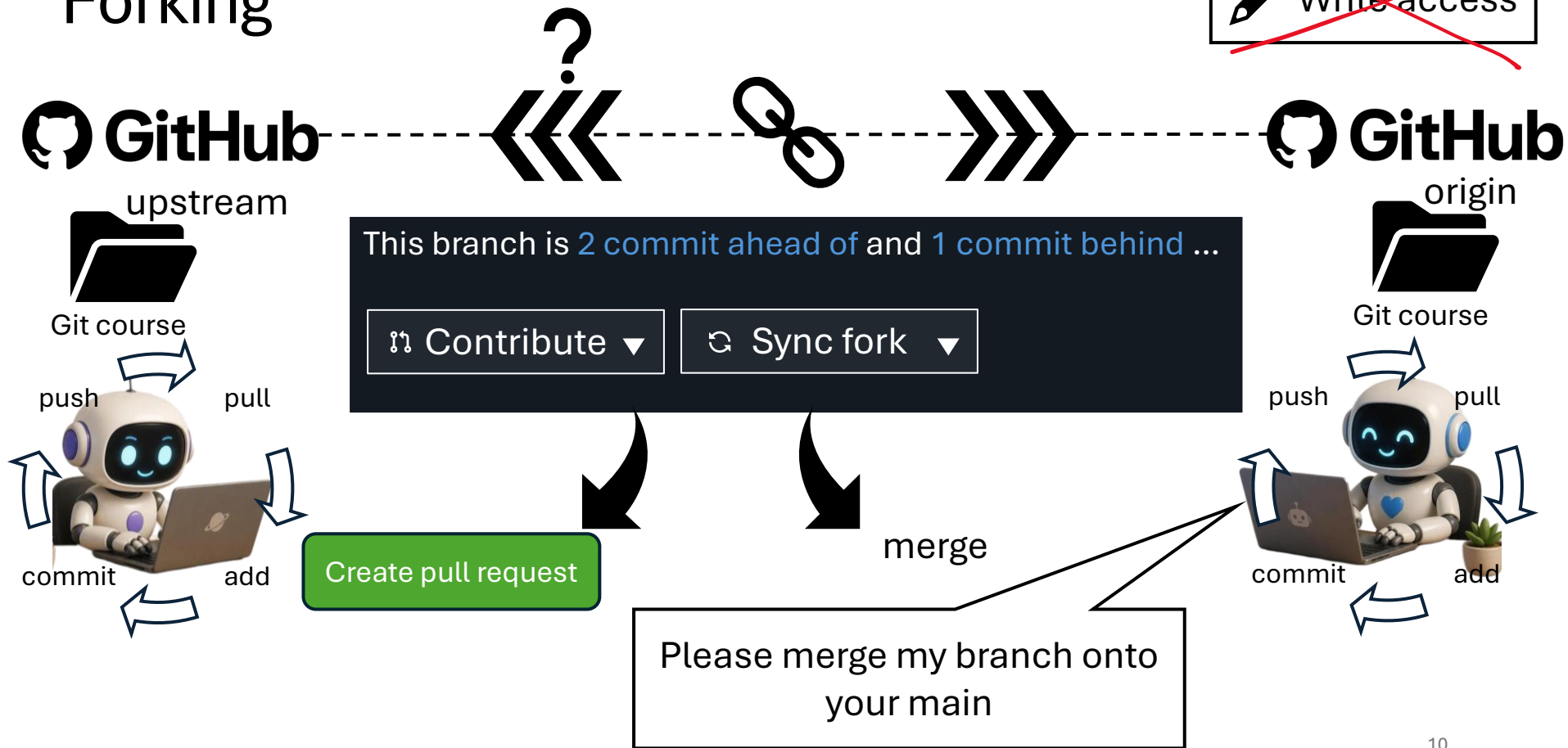
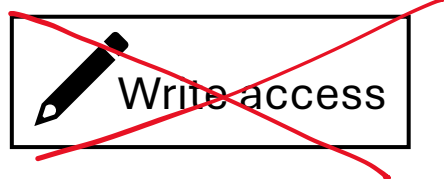
Git course
upstream



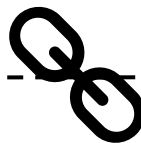
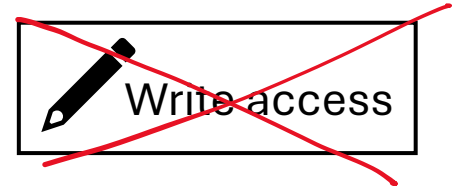
Git course
origin



Forking



Creating a pull requests



✓ Able to merge

Title: New analysis

Description: replace ANOVA with Mixed effects model

Changes:

Create pull request

Standard ANOVA

```
23 - mod_aov <- aov(react ~ tays, data = sleep)
```

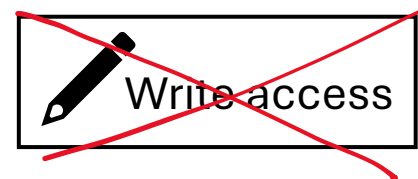
```
23 + mod_lmer <- lmer(react ~ days + (1 + days | sub), data = sleep)
```



Git course



Review and merge a pull requests



Git course



Check commits and changes

Standard ANOVA

```
23 - mod_aov <- aov(react ~ tays, data = sleep)
```

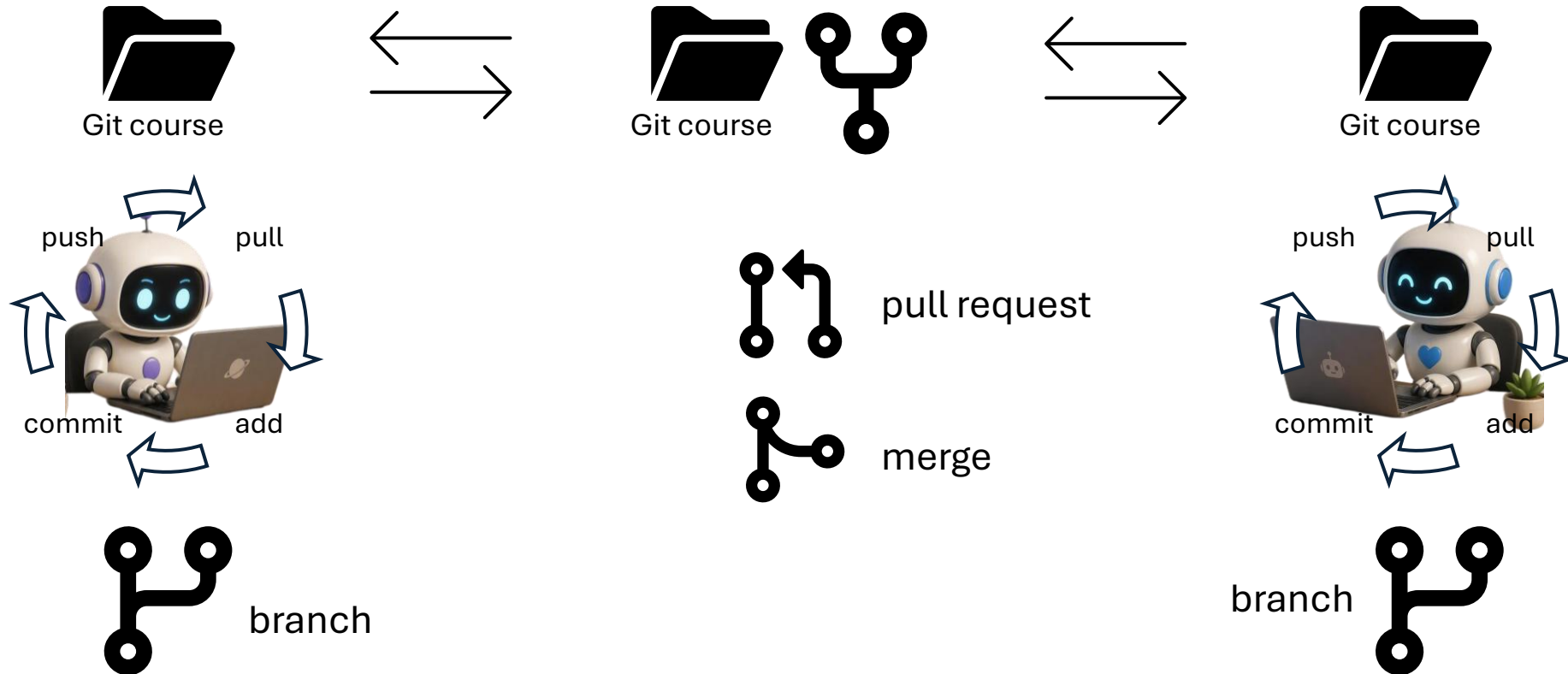
```
23 + mod_lmer <- lmer(react ~ days + (1 + days | sub), data = sleep)
```

Comment: Looks good!



Merge pull request

Collaboration



Summary

- Cloning inherits the remote for you: `origin` is wired up automatically, unlike `git init` where you add it by hand.
- `fetch` downloads without touching your files; `pull` is `fetch + merge` in one step.
- Forking makes your own remote copy on GitHub; you clone the fork, so `origin` is *yours*.
- `upstream` is the conventional name for the original project's remote, added with `git remote add`.
- A pull request asks a project to merge your branch; merging it is a normal Git merge triggered from GitHub, and you still need to `pull` afterwards to see it locally.